

(1) A chemical reaction $A \rightarrow B$ is carried out in a closed vessel. The following data are taken for the concentration of A, $C_A(g/L)$, as a function of time, $t(min)$, from the start of the reaction:

$t(min)$	0	36	65	100	160	∞
$C_A(g/L)$	0.1823	0.1453	0.1216	0.1025	0.0795	0.0495

A proposed reaction mechanism predicts that C_A and t should be related by the expression

$$\ln \frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = -kt$$

where k is a reaction rate constant, and the zero and infinity subscripts on the concentrations indicate the initial and very long time concentration values.

- Do the data support this prediction? If so, determine the value of k .
- If the tank volume is 30.5 gallons and there is no B in the tank at $t = 0$, how much B in grams does the tank contain in two hours?

(2) (a) Find two different ways to linearize the equation $r = Az/(B + z)$ where A and B are adjustable constants. The reaction rate is r , and the concentration is z . Use both linearizations to fit the data to the right using perpendicular offsets. Compare the values of the constants for the two linearizations and the quality Q^2 of the fits.

(b) Use the solver on Excel to find a numerical solution for fitting the nonlinear equation to the data using vertical offsets. You will need a reasonable first guess for the solver in order to get the numerical method to converge. Use one of your results in part (a) for this guess. Compare this result with the two results in part (a).

z	r
2.50E+00	20
4.00E+00	43
1.00E+01	65
2.00E+01	106
4.00E+01	112
1.00E+02	128
2.00E+02	150
1.00E+03	165

(3) Murphy 1.12

(4) Murphy 2.11 (1 ton = 2000 lb_m)

(5) A mixture of ethanol and water contains 60 wt% water.

- Assuming volume additivity of the components, estimate the specific gravity of the mixture at 20°C. What volume (in liters) of this mixture is required to provide 150 moles of ethanol?
- Repeat part (a) with the additional information that the specific gravity of the mixture at 20°C is 0.93518 (making it unnecessary to assume volume additivity). What percentage error for the volume of the mixture arises from the volume additivity assumption?

1.A. $\ln \frac{C_A - C_{A\infty}}{C_{A0} - C_{A\infty}} = -kt$

$$\ln \frac{(0.1823 - 0.0495)}{(0.1823 - 0.0495)} = -k(0) \Rightarrow k = 0$$

$$\ln \frac{(0.1453 - 0.0495)}{(0.1823 - 0.0495)} = -k(36) \Rightarrow k = 9.16E-3$$

$$\ln \frac{(0.1216 - 0.0495)}{(0.1823 - 0.0495)} = -k(65) \Rightarrow k = 9.40E-3$$

$$\ln \frac{(0.1025 - 0.0495)}{(0.1823 - 0.0495)} = -k(100) \Rightarrow k = 9.19E-3$$

$$\ln \frac{(0.0795 - 0.0495)}{(0.1823 - 0.0495)} = -k(160) \Rightarrow k = 9.30E-3$$

$$k = \frac{[(9.16E-3) + (9.4E-3) + (9.19E-3) + (9.3E-3)]}{4}$$

$k = 9.26E-3$



YES! The data supports this prediction.
This can be proven by k's consistency within the data values.

1.b. $V = 30.5$ gallons $B(0) = 0$ $B(2\text{hr}) = ? \text{ g}$

$$\ln\left(\frac{C_A - 0.0495}{0.1823 - 0.0495}\right) = -(9.26 \text{E-}3)(120)$$

$$\ln\left(\frac{C_A - 0.0495}{0.1823 - 0.0495}\right) = -1.11$$

$$\frac{C_A - 0.0495}{(132.8 \text{E-}3)} = 329.17 \text{E-}3$$

$$C_A - 0.0495 = 43.71 \text{E-}3$$

$$C_A = 93.21 \text{E-}3 \frac{\text{g}}{\text{L}}$$

$$30.5 \text{ gallons} = 115.46 \text{ L}$$

$$93.21 \text{E-}3 \frac{\text{g}}{\text{L}} \times 115.46 \text{ L} = \boxed{10.76 \text{ g}}$$

20/20



2.a.

$$r = \frac{Az}{B+Z} \Rightarrow \frac{1}{r} = \frac{B}{A} \frac{1}{Z} + \frac{1}{A}$$

$$(y = mx + b)$$



$$y = \frac{1}{r} \quad x = \frac{1}{Z} \quad \text{slope} = \frac{B}{A} \quad \text{intercept} = \frac{1}{A}$$

$$r = \frac{Az}{B+z}$$

$$r(B+z) = Az$$

$$rB + rz = Az$$

$$B + z = \frac{Az}{r}$$

$$\frac{B}{A} + \frac{z}{A} = \frac{z}{r} \Rightarrow \frac{z}{r} = \frac{1}{A}z + \frac{B}{A}$$

$$(y = mx + b)$$

$$y = \frac{z}{r} \quad x = z \quad \text{slope} = \frac{1}{A} \quad \text{intercept} = \frac{B}{A}$$

1st Linearization

2nd Linearization

A: 184.424

A: 165.961

B: 18.471

B: 18.026

Q²: 0.9504

Q² = 0.9999

The value B are very similar for both linearizations. The value of A differ significantly, the first one being much larger than the second. Lastly, the 2nd linearization indicates a significantly

better fit with a Q^2 closer to 1.

2.b.

$$A: 157.99$$

$$B: 13.23$$

$$Q^2: 0.9765$$



no extra
file.
-15

Values A & B are much smaller than the values from part A. This is expected because linearization alters error structure.

10/25

3. Murphy 1.12

$$\text{Price}_{\text{glucose}} = 43,300 - 150 - 80,850P$$

$$43,300 - 150 - 80,850P = 27,100$$

$$P = \frac{43300 - 150 - 27100}{80850} = 0.1985 \approx 20 \text{ \$/kg}$$

$$0.6 - 0.1985 = 0.4015 \approx 0.40 \text{ \$/kg}$$

X - 5

$$\text{Price}_{\text{glucose}} = 97700 - 80770P$$

$$97700 - 80770P = 89300$$

$$P = \frac{97700 - 89300}{80770} = 0.1040 \approx 0.10 \text{ \$/kg}$$

10/20

$$0.6 - 0.1040 = 0.496 \approx 0.5 \text{ \$/kg}$$

X-5

4. Murphy 2.11 (1 ton = 2000 lbm)

1970:

$$14 \text{ g/mile}, 1100 \text{ billion} = 1.1 \times 10^{12} \text{ miles}$$

2000:

$$1 \text{ g/mile}, 2600 \text{ billion} = 2.6 \times 10^{12} \text{ miles}$$

15/15

$$14 \frac{\text{g}}{\text{mile}} \times 1.1 \times 10^{12} \text{ miles} = 1.54 \times 10^{13} \text{ g} \div 9.07 \times 10^5 \frac{\text{g}}{\text{ton}} = 1.70 \times 10^7 \text{ tons}$$

$$1 \frac{\text{g}}{\text{mile}} \times 2.6 \times 10^{12} \text{ miles} = 2.6 \times 10^{12} \text{ g} \div 9.07 \times 10^5 \frac{\text{g}}{\text{ton}} = 2.87 \times 10^6 \text{ tons}$$

$$\% \text{ change} = \frac{(2.87 \times 10^6 - 1.7 \times 10^7)}{(1.7 \times 10^7)} \times 100\%$$

$$= 83\% \text{ decrease over 30 years.}$$

5.A $\frac{60g H_2O}{100g Sol}$ Specific gravity?

40% Ethanol: $\frac{40g Ethanol}{100g Solution}$

$$\frac{60g}{1 \frac{g}{mL}} + \frac{40g}{0.789 \frac{g}{mL}} = 110.7 mL$$

20/20

$$\rho = \frac{100g}{110.7mL} = \boxed{0.903 \frac{g}{mL}}$$

$$150 mol Ethanol \times \frac{46.07 g Ethanol}{1 mol Ethanol} = 6910.5 g Ethanol$$

$$6910.2 g Ethanol \times \frac{100g Solution}{40g Ethanol} = 17276.3 g Sol$$

$$17276.3 g Sol \times \frac{1 mL}{0.903 g} \times \frac{10^{-3} L}{1 mL} = 19.13 L$$

5.b.

$$17276.3 g Sol \times \frac{1 mL}{0.93518 g} \times \frac{10^{-3} L}{1 mL} = 18.47 L$$

$$\frac{(19.13 - 18.47) L}{(18.47 L)} \times 100\% = \boxed{3.57\% \text{ error}}$$